Tribhuvan University
Institute of Science and Technology
2079
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Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology (MTH 112)
(Mathematics I)
(NEW COURSE)

Full Marks: 80
Pass Marks: 32
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
The figures in the margin indicate full marks.

Group A ($10 \times 3 = 30$)

Attempt any **THREE** questions.

1. (a) If a function is defined by [1+1+1+2]

$$f(x) = \begin{cases} 1+x, \text{ if } x \leq -1 \\ x^2, \text{ if } x > -1, \end{cases}$$

evaluate $f(-3), f(-1)$ and $f(0)$ and sketch the graph.

(b) Prove that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist. [5]

2. (a) Sketch the curve $y = x^2 + 1$ with the guidelines of sketching [5]

(b) If $z = xy^2 + y^3, x = \sin t, y = \cos t$, find $\frac{dz}{dt}$ at $t = 0$. [5]

3. (a) Estimate the area between the curve $y = x^2$ and the lines $x = 0$ and $x = 1$, using rectangle method, with four sub intervals. [5]

(b) A particle moves along a line so that its velocity v at time t is [5]

$$v = t^2 - 2t + 10$$

(i) Find the displacement of the particle during the time period $1 \leq t \leq 4$.
(ii) Find the distance traveled during this time period.

4. (a) Define initial value problem. Solve: [5]

$$y'' + y' - 6y = 0, y(0) = 1, y'(0) = 0$$

(b) Find the Taylor's series expansion for $\cos x$ at $x = 0$. [5]

Group B ($10 \times 5 = 50$)

Attempt any **TEN** questions.

5. (a) Dry air is moving upward. If the ground temperature is 20° and the temperature at a height of $2km$ is 10^oC, express the temperature T in ^oC as a function of the height h (in kilometers), assuming that a linear model is appropriate. (b) Draw the graph of the function and find the slope. Hence, give the meaning of the slope. (c) What is the temperature at a height of $2km$? [5]

6. Find the equation of tangent at $(1, 3)$ to the curve $y = 2x^2 + 1$.

7. State Rolle's theorem and verify the theorem for $f(x) = x^2 - 9, x \in [-3, 3]$.

8. Starting with $x_1 = 1$, find the third approximation x_3 to the root of the equation $x^3 - x - 5 = 0$.

9. Show that the integral

$$\int_0^3 \frac{dx}{x-1},$$

diverges.

10. Use Trapezoidal rule to approximate the integral $\int_1^2 \frac{dx}{x}$, with $n = 5$.

11. Find the derivative of $\mathbf{r}(t) = t^2\mathbf{i} - te^{-t}\mathbf{j} + \sin 2t\mathbf{k}$ and find the unit tangent vector at $t = 0$.

12. What is a sequence? Is the sequence

$$a_n = \frac{n}{5+n}$$

convergent?

13. Find the angle between the vectors $\mathbf{a} = (2, 2, -1)$ and $\mathbf{b} = (1, 3, 2)$.

14. Find the partial derivative f_{xx} and f_{yy} of $f(x, y) = x^2 + x^3y^2 - y^2 + xy$, at $(1, 2)$.

15. Evaluate

(a) $\int_0^3 \int_1^2 x^2y \, dxdy$

(b) $\int_0^\pi \int_1^2 y \sin(xy) dxdy$